

Question ID: 4b7bb316

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

The length of each edge of a box is **29** inches. Each side of the box is in the shape of a square. The box does not have a lid. What is the exterior surface area, in square inches, of this box without a lid?

Correct Answer: 4205

Rationale

The correct answer is **4,205**. The exterior surface area of a figure is the sum of the areas of all its faces. It's given that the box does not have a lid and that each side of the box is in the shape of a square. Therefore, the box consists of **5** congruent square faces. It's also given that the length of each edge is **29** inches. Let *s* represent the length of an edge of a square. It follows that the area of a square is equal to *s*². Therefore, the area of each of the **5** square faces is equal to **29²**, or **841**, square inches. Since the box consists of **5** congruent square faces, it follows that the sum of the areas of all its faces, or the exterior surface area of this box without a lid, is **5(841)**, or **4,205**, square inches.